

## 2 KNN Prediction

- Do this manually (no code) and show your work. You can use the following table or insert a photo. If there are ties, indicate so and use random selection to break them.

The table below provides a training data set of six observations of three predictors, and one qualitative response variable  $Y$ .

Predict for  $Y$  when  $X_1 = 1, X_2 = 0, X_3 = 1$  using K-nearest neighbors.

- a. Compute the Euclidean distance between each observation and the test point,  $X_1 = 1, X_2 = 0, X_3 = 1$ .

- b. What is your prediction with  $k = 1$ ? Why? It's "Red" because the nearest neighbor had  $Y = \text{Red}$  - observation 6.

- c. What is your prediction with  $k = 3$ ? Why? It's "Green" because based on the 3 closest neighbors distance, we observed: Green = 2 and Red = 1,  $\frac{2}{3} > \frac{1}{3} \therefore \hat{Y} = \text{GREEN}$

| Obs.                  | $X_1$ | $X_2$ | $X_3$ | $Y$   | Dist to<br>$\vec{X} = (1, 0, 1)$                                     | Neigh<br>for<br>$k = 1$ | Neigh<br>for<br>$k = 3$ |
|-----------------------|-------|-------|-------|-------|----------------------------------------------------------------------|-------------------------|-------------------------|
| 1                     | 0     | 3     | 0     | Red   | $D = \sqrt{(0-1)^2 + (3-0)^2 + (0-1)^2}$<br>$D = \sqrt{11} = (3.32)$ |                         |                         |
| 2                     | 2     | 0     | 0     | Green | $D = \sqrt{(2-1)^2 + (0-0)^2 + (0-1)^2}$<br>$D = \sqrt{2} = (1.41)$  |                         | Green                   |
| 3                     | 0     | 1     | 3     | Red   | $D = \sqrt{(0-1)^2 + (1-0)^2 + (3-1)^2}$<br>$D = \sqrt{6} = (2.45)$  |                         |                         |
| 4                     | 0     | 1     | 2     | Green | $D = \sqrt{(0-1)^2 + (1-0)^2 + (2-1)^2}$<br>$D = \sqrt{3} = (1.73)$  |                         | Green                   |
| 5                     | -1    | 0     | 1     | Green | $D = \sqrt{(-1-1)^2 + (0-0)^2 + (1-1)^2}$<br>$D = \sqrt{4} = (2)$    |                         |                         |
| 6                     | 1     | 1     | 1     | Red   | $D = \sqrt{(1-1)^2 + (1-0)^2 + (1-1)^2}$<br>$D = \sqrt{1} = (1)$     | Red                     | Red                     |
| $\hat{Y} \Rightarrow$ |       |       |       |       |                                                                      | Red                     | Green                   |

- d. In classification problems, decision boundaries separate different response categories in the  $\vec{X}$  space of predictor variables. If the decision boundary in this problem is highly nonlinear, would we expect the best value for  $k$  to be large or small? Why?

For a highly nonlinear model, a small value of  $k$  would be best because a small  $k$  is more flexible and can model the data more closely allowing for low bias, but high variance.